

CCNA 1 v7.0 - Introduction to Networks

Final Exam Practice Quiz — Questions with Correct Answers Marked

111 questions. Correct answer(s) are highlighted in green and marked with a checkmark (✓).

Q1. Which two traffic types use the Real-Time Transport Protocol (RTP)? (Choose two.)

✓ A. video

B. web

C. file transfer

✓ D. voice

E. peer to peer

Q2. Which wireless technology has low-power and data rate requirements making it popular in home automation applications?

✓ A. ZigBee

B. LoRaWAN

C. 5G

D. Wi-Fi

Q3. Which layer of the TCP/IP model provides a route to forward messages through an internetwork?

A. application

B. network access

✓ C. internet

D. transport

Q4. Which type of server relies on record types such as A, NS, AAAA, and MX in order to provide services?

✓ A. DNS

B. email

C. file

D. web

Q5. What are proprietary protocols?

A. protocols developed by private organizations to operate on any vendor hardware

B. protocols that can be freely used by any organization or vendor

✓ C. protocols developed by organizations who have control over their definition and operation

D. a collection of protocols known as the TCP/IP protocol suite

Q6. What service is provided by DNS?

✓ A. Resolves domain names, such as cisco.com, into IP addresses.

B. A basic set of rules for exchanging text, graphic images, sound, video, and other multimedia files on the web.

C. Allows for data transfers between a client and a file server.

D. Uses encryption to secure the exchange of text, graphic images, sound, and video on the web.

Q7. A client packet is received by a server. The packet has a destination port number of 110. What service is the client requesting?

A. DNS

B. DHCP

C. SMTP

✓ D. POP3

Q8. What command can be used on a Windows PC to see the IP configuration of that computer?

A. show ip interface brief

B. ping

C. show interfaces

✓ D. ipconfig

Q9. A wired laser printer is attached to a home computer. That printer has been shared so that other computers on the home network can also use the printer. What networking model is in use?

A. client-based

B. master-slave

C. point-to-point

✓ D. peer-to-peer (P2P)

Q10. What characteristic describes a virus?

A. a network device that filters access and traffic coming into a network

B. the use of stolen credentials to access private data

C. an attack that slows or crashes a device or network service

✓ D. malicious software or code running on an end device

Q11. Three bank employees are using the corporate network. The first employee uses a web browser to view a company web page in order to read some announcements. The

second employee accesses the corporate database to perform some financial transactions. The third employee participates in an important live audio conference with other corporate managers in branch offices. If QoS is implemented on this network, what will be the priorities from highest to lowest of the different data types?

- A. financial transactions, web page, audio conference
- ✓ B. audio conference, financial transactions, web page
- C. financial transactions, audio conference, web page
- D. audio conference, web page, financial transactions

Q13. Refer to the exhibit. If Host1 were to transfer a file to the server, what layers of the TCP/IP model would be used?

- A. only application and Internet layers
- B. only Internet and network access layers
- C. only application, Internet, and network access layers
- ✓ D. application, transport, Internet, and network access layers
- E. only application, transport, network, data link, and physical layers
- F. application, session, transport, network, data link, and physical layers

Q15. Refer to the exhibit. The IP address of which device interface should be used as the default gateway setting of host H1?

- A. R1: S0/0/0
- B. R2: S0/0/1
- ✓ C. R1: G0/0
- D. R2: S0/0/0

Q16. What service is provided by Internet Messenger?

- ✓ A. An application that allows real-time chatting among remote users.
- B. Allows remote access to network devices and servers.
- C. Resolves domain names, such as cisco.com, into IP addresses.
- D. Uses encryption to provide secure remote access to network devices and servers.

Q17. Refer to the exhibit. Match the network with the correct IP address and prefix that will satisfy the usable host addressing requirements for each network. (Network A=192.168.0.128/25, B=192.168.0.0/26, C=192.168.0.96/27, D=192.168.0.80/30)

- ✓ A. Network A: 192.168.0.128 /25
- ✓ B. Network B: 192.168.0.0 /26
- ✓ C. Network C: 192.168.0.96 /27
- ✓ D. Network D: 192.168.0.80 /30

Q18. Refer to the exhibit. Which protocol was responsible for building the table that is shown? (output of arp -a)

- A. DHCP
- ✓ B. ARP
- C. DNS
- D. ICMP

Q19. A network administrator notices that some newly installed Ethernet cabling is carrying corrupt and distorted data signals. The new cabling was installed in the ceiling close to fluorescent lights and electrical equipment. Which two factors may interfere with the copper cabling and result in signal distortion and data corruption? (Choose two.)

- A. crosstalk
- B. extended length of cabling
- ✓ C. RFI
- ✓ D. EMI
- E. signal attenuation

Q20. A host is trying to send a packet to a device on a remote LAN segment, but there are currently no mappings in its ARP cache. How will the device obtain a destination MAC address?

- A. It will send the frame and use its own MAC address as the destination.
- B. It will send an ARP request for the MAC address of the destination device.
- C. It will send the frame with a broadcast MAC address.
- D. It will send a request to the DNS server for the destination MAC address.
- ✓ E. It will send an ARP request for the MAC address of the default gateway.

Q22. A client packet is received by a server. The packet has a destination port number of 53. What service is the client requesting?

- ✓ A. DNS
- B. NetBIOS (NetBT)
- C. POP3
- D. IMAP

Q23. A network administrator is adding a new LAN to a branch office. The new LAN must support 25 connected devices. What is the smallest network mask that the network administrator can use for the new network?

- A. 255.255.255.128
- B. 255.255.255.192
- ✓ C. 255.255.255.224

D. 255.255.255.240

Q24. What characteristic describes a Trojan horse?

- ✓ A. malicious software or code running on an end device
- B. an attack that slows or crashes a device or network service
- C. the use of stolen credentials to access private data
- D. a network device that filters access and traffic coming into a network

Q25. What service is provided by HTTPS?

- A. Uses encryption to provide secure remote access to network devices and servers.
- B. Resolves domain names, such as cisco.com, into IP addresses.
- ✓ C. Uses encryption to secure the exchange of text, graphic images, sound, and video on the web.
- D. Allows remote access to network devices and servers.

Q26. A technician with a PC is using multiple applications while connected to the Internet. How is the PC able to keep track of the data flow between multiple application sessions and have each application receive the correct packet flows?

- A. The data flow is being tracked based on the destination MAC address of the technician PC.
- ✓ B. The data flow is being tracked based on the source port number that is used by each application.
- C. The data flow is being tracked based on the source IP address that is used by the PC of the technician.
- D. The data flow is being tracked based on the destination IP address that is used by the PC of the technician.

Q27. A network administrator is adding a new LAN to a branch office. The new LAN must support 61 connected devices. What is the smallest network mask that the network administrator can use for the new network?

- A. 255.255.255.240
- B. 255.255.255.224
- ✓ C. 255.255.255.192
- D. 255.255.255.128

Q28. Refer to the exhibit. Match the network with the correct IP address and prefix. (A=192.168.0.0/25, B=192.168.0.128/26, C=192.168.0.192/27, D=192.168.0.224/30)

- ✓ A. Network A: 192.168.0.0 /25
- ✓ B. Network B: 192.168.0.128 /26
- ✓ C. Network C: 192.168.0.192 /27
- ✓ D. Network D: 192.168.0.224 /30

Q29. What characteristic describes a DoS attack?

- A. the use of stolen credentials to access private data
- B. a network device that filters access and traffic coming into a network
- C. software that is installed on a user device and collects information about the user
- ✓ D. an attack that slows or crashes a device or network service

Q31. What service is provided by SMTP?

- ✓ A. Allows clients to send email to a mail server and the servers to send email to other servers.
- B. Allows remote access to network devices and servers.
- C. Uses encryption to provide secure remote access to network devices and servers.
- D. An application that allows real-time chatting among remote users.

Q32. Which scenario describes a function provided by the transport layer?

- A. A student is using a classroom VoIP phone to call home. The unique identifier burned into the phone is a transport layer address used to contact another network device on the same network.
- B. A student is playing a short web-based movie with sound. The movie and sound are encoded within the transport layer header.
- ✓ C. A student has two web browser windows open in order to access two web sites. The transport layer ensures the correct web page is delivered to the correct browser window.
- D. A corporate worker is accessing a web server located on a corporate network. The transport layer formats the screen so the web page appears properly no matter what device is being used to view the web site.

Q33. Refer to the exhibit. Host B on subnet Teachers transmits a packet to host D on subnet Students. Which Layer 2 and Layer 3 addresses are contained in the PDUs that are transmitted from host B to the router?

- ✓ A. L2 dest=00-00-0c-94-36-ab, L2 src=00-00-0c-94-36-bb, L3 dest=172.16.20.200, L3 src=172.16.10.200
- B. L2 dest=00-00-0c-94-36-dd, L2 src=00-00-0c-94-36-bb, L3 dest=172.16.20.200, L3 src=172.16.10.200
- C. L2 dest=00-00-0c-94-36-cd, L2 src=00-00-0c-94-36-bb, L3 dest=172.16.20.99, L3 src=172.16.10.200
- D. L2 dest=00-00-0c-94-36-ab, L2 src=00-00-0c-94-36-bb, L3 dest=172.16.20.200, L3 src=172.16.100.200

Q34. What does the term "attenuation" mean in data communication?

- A. strengthening of a signal by a networking device
- B. leakage of signals from one cable pair to another
- C. time for a signal to reach its destination

✓ D. loss of signal strength as distance increases

Q35. Refer to the exhibit. An administrator is trying to configure the switch but receives the error message that is displayed in the exhibit. What is the problem?

A. The entire command, configure terminal, must be used.

B. The administrator is already in global configuration mode.

✓ C. The administrator must first enter privileged EXEC mode before issuing the command.

D. The administrator must connect via the console port to access global configuration mode.

Q36. Which two protocols operate at the top layer of the TCP/IP protocol suite? (Choose two.)

A. TCP

B. IP

C. UDP

✓ D. POP

✓ E. DNS

F. Ethernet

Q37. A company has a file server that shares a folder named Public. The network security policy specifies that the Public folder is assigned Read-Only rights to anyone who can log into the server while the Edit rights are assigned only to the network admin group. Which component is addressed in the AAA network service framework?

A. automation

B. accounting

C. authentication

✓ D. authorization

Q38. What three requirements are defined by the protocols used in network communications to allow message transmission across a network? (Choose three.)

✓ A. message size

✓ B. message encoding

C. connector specifications

D. media selection

✓ E. delivery options

F. end-device installation

Q39. What are two characteristics of IP? (Choose two.)

✓ A. does not require a dedicated end-to-end connection

✓ B. operates independently of the network media

- C. retransmits packets if errors occur
- D. re-assembles out of order packets into the correct order at the receiver end
- E. guarantees delivery of packets

Q40. An employee remotely logs into the company using the appropriate username and password to attend a video conference. The connection to the ISP failed but a secondary connection activated within seconds, unnoticed. What three network characteristics are described in this scenario? (Choose three.)

- ✓ A. security
- ✓ B. quality of service
- C. scalability
- D. powerline networking
- E. integrity
- ✓ F. fault tolerance

Q41. What are two common causes of signal degradation when using UTP cabling? (Choose two.)

- ✓ A. improper termination
- B. low-quality shielding in cable
- C. installing cables in conduit
- ✓ D. low-quality cable or connectors
- E. loss of light over long distances

Q42. Which subnet would include the address 192.168.1.96 as a usable host address?

- ✓ A. 192.168.1.64/26
- B. 192.168.1.32/27
- C. 192.168.1.32/28
- D. 192.168.1.64/29

Q43. Refer to the exhibit. On the basis of the output, which two statements about network connectivity are correct? (Choose two.)

- A. This host does not have a default gateway configured.
- ✓ B. There are 4 hops between this device and the device at 192.168.100.1.
- ✓ C. There is connectivity between this device and the device at 192.168.100.1.
- D. The connectivity between these two hosts allows for videoconferencing calls.
- E. The average transmission time between the two hosts is 2 milliseconds.

Q44. Which two statements describe how to assess traffic flow patterns and network traffic types using a protocol analyzer? (Choose two.)

A. Capture traffic on the weekends when most employees are off work.

✓ B. Capture traffic during peak utilization times to get a good representation of the different traffic types.

C. Only capture traffic in the areas of the network that receive most of the traffic such as the data center.

✓ D. Perform the capture on different network segments.

E. Only capture WAN traffic because traffic to the web is responsible for the largest amount of traffic on a network.

Q45. What is the consequence of configuring a router with the ipv6 unicast-routing global configuration command?

A. All router interfaces will be automatically activated.

✓ B. The IPv6 enabled router interfaces begin sending ICMPv6 Router Advertisement messages.

C. Each router interface will generate an IPv6 link-local address.

D. It statically creates a global unicast address on this router.

Q46. Which three layers of the OSI model map to the application layer of the TCP/IP model? (Choose three.)

✓ A. application

B. network

C. data link

✓ D. session

✓ E. presentation

F. transport

Q47. Refer to the exhibit. If PC1 is sending a packet to PC2 and routing has been configured between the two routers, what will R1 do with the Ethernet frame header attached by PC1?

A. nothing, because the router has a route to the destination network

B. open the header and use it to determine whether the data is to be sent out S0/0/0

C. open the header and replace the destination MAC address with a new one

✓ D. remove the Ethernet header and configure a new Layer 2 header before sending it out S0/0/0

Q48. What will happen if the default gateway address is incorrectly configured on a host?

A. The host cannot communicate with other hosts in the local network.

✓ B. The host cannot communicate with hosts in other networks.

C. A ping from the host to 127.0.0.1 would not be successful.

D. The host will have to use ARP to determine the correct address of the default gateway.

E. The switch will not forward packets initiated by the host.

Q49. What are two features of ARP? (Choose two.)

A. When a host is encapsulating a packet into a frame, it refers to the MAC address table to determine the mapping of IP addresses to MAC addresses.

B. An ARP request is sent to all devices on the Ethernet LAN and contains the IP address of the destination host and its multicast MAC address.

✓ C. If a host is ready to send a packet to a local destination device and it has the IP address but not the MAC address of the destination, it generates an ARP broadcast.

D. If no device responds to the ARP request, then the originating node will broadcast the data packet to all devices on the network segment.

✓ E. If a device receiving an ARP request has the destination IPv4 address, it responds with an ARP reply.

Q50. A network administrator is adding a new LAN to a branch office. The new LAN must support 90 connected devices. What is the smallest network mask that the network administrator can use for the new network?

✓ A. 255.255.255.128

B. 255.255.255.240

C. 255.255.255.248

D. 255.255.255.224

Q51. What are two ICMPv6 messages that are not present in ICMP for IPv4? (Choose two.)

✓ A. Neighbor Solicitation

B. Destination Unreachable

C. Host Confirmation

D. Time Exceeded

✓ E. Router Advertisement

F. Route Redirection

Q52. A client packet is received by a server. The packet has a destination port number of 80. What service is the client requesting?

A. DHCP

B. SMTP

C. DNS

✓ D. HTTP

Q53. What is an advantage for small organizations of adopting IMAP instead of POP?

A. POP only allows the client to store messages in a centralized way, while IMAP allows distributed storage.

✓ **B. Messages are kept in the mail servers until they are manually deleted from the email client.**

C. When the user connects to a POP server, copies of the messages are kept in the mail server for a short time, but IMAP keeps them for a long time.

D. IMAP sends and retrieves email, but POP only retrieves email.

Q54. A technician can ping the IP address of the web server of a remote company but cannot successfully ping the URL address of the same web server. Which software utility can the technician use to diagnose the problem?

A. tracer

B. ipconfig

C. netstat

✓ **D. nslookup**

Q55. Which two functions are performed at the LLC sublayer of the OSI Data Link Layer to facilitate Ethernet communication? (Choose two.)

A. implements CSMA/CD over legacy shared half-duplex media

✓ **B. enables IPv4 and IPv6 to utilize the same physical medium**

C. integrates Layer 2 flows between 10 Gigabit Ethernet over fiber and 1 Gigabit Ethernet over copper

D. implements a process to delimit fields within an Ethernet 2 frame

✓ **E. places information in the Ethernet frame that identifies which network layer protocol is being encapsulated by the frame**

Q56. The global configuration command ip default-gateway 172.16.100.1 is applied to a switch. What is the effect of this command?

A. The switch can communicate with other hosts on the 172.16.100.0 network.

✓ **B. The switch can be remotely managed from a host on another network.**

C. The switch is limited to sending and receiving frames to and from the gateway 172.16.100.1.

D. The switch will have a management interface with the address 172.16.100.1.

Q57. What happens when the transport input ssh command is entered on the switch vty lines?

A. The SSH client on the switch is enabled.

B. The switch requires a username/password combination for remote access.

✓ **C. Communication between the switch and remote users is encrypted.**

D. The switch requires remote connections via a proprietary client software.

**Q58. Match the type of threat with the cause.
(electrical/hardware/environmental/maintenance)**

✓ **A. electrical threats -> voltage spikes, insufficient supply voltage (brownouts), unconditioned power (noise), and total power loss**

✓ B. hardware threats -> physical damage to servers, routers, switches, cabling plant, and workstations

✓ C. environmental threats -> temperature extremes or humidity extremes

✓ D. maintenance threats -> poor handling of key electrical components (ESD), lack of critical spare parts, poor cabling, and poor labeling

Q59. A disgruntled employee is using some free wireless networking tools to determine information about the enterprise wireless networks. This person is planning on using this information to hack the wireless network. What type of attack is this?

A. DoS

B. access

✓ C. reconnaissance

D. Trojan horse

Q60. What service is provided by HTTP?

A. Uses encryption to secure the exchange of text, graphic images, sound, and video on the web.

B. Allows for data transfers between a client and a file server.

C. An application that allows real-time chatting among remote users.

✓ D. A basic set of rules for exchanging text, graphic images, sound, video, and other multimedia files on the web.

Q61. A client packet is received by a server. The packet has a destination port number of 67. What service is the client requesting?

A. FTP

✓ B. DHCP

C. Telnet

D. SSH

Q62. What are two problems that can be caused by a large number of ARP request and reply messages? (Choose two.)

A. Switches become overloaded because they concentrate all the traffic from the attached subnets.

✓ B. The ARP request is sent as a broadcast, and will flood the entire subnet.

C. The network may become overloaded because ARP reply messages have a very large payload due to the 48-bit MAC address and 32-bit IP address that they contain.

D. A large number of ARP request and reply messages may slow down the switching process, leading the switch to make many changes in its MAC table.

✓ E. All ARP request messages must be processed by all nodes on the local network.

Q63. A group of Windows PCs in a new subnet can access local network resources but not the Internet. Which three Windows CLI commands and utilities will provide the

necessary information? (Choose three.)

- A. netsh interface ipv6 show neighbor
- B. arp -a
- C. tracert
- ✓ D. ping
- ✓ E. ipconfig
- ✓ F. nslookup
- G. telnet

Q64. During the process of forwarding traffic, what will the router do immediately after matching the destination IP address to a network on a directly connected routing table entry?

- A. analyze the destination IP address
- ✓ B. switch the packet to the directly connected interface
- C. look up the next-hop address for the packet
- D. discard the traffic after consulting the route table

Q65. What characteristic describes antispyware?

- ✓ A. applications that protect end devices from becoming infected with malicious software
- B. a network device that filters access and traffic coming into a network
- C. software on a router that filters traffic based on IP addresses or applications
- D. a tunneling protocol that provides remote users with secure access into the network of an organization

Q66. A network administrator needs to keep the user ID, password, and session contents private when establishing remote CLI connectivity with a switch to manage it. Which access method should be chosen?

- A. Telnet
- B. AUX
- ✓ C. SSH
- D. Console

Q67. What are the two most effective ways to defend against malware? (Choose two.)

- A. Implement a VPN.
- B. Implement network firewalls.
- C. Implement RAID.
- D. Implement strong passwords.
- ✓ E. Update the operating system and other application software.
- ✓ F. Install and update antivirus software.

Q68. Which type of security threat would be responsible if a spreadsheet add-on disables the local software firewall?

- A. brute-force attack
- ✓ B. Trojan horse
- C. DoS
- D. buffer overflow

Q69. Which frame field is created by a source node and used by a destination node to ensure that a transmitted data signal has not been altered by interference, distortion, or signal loss?

- A. User Datagram Protocol field
- B. transport layer error check field
- C. flow control field
- ✓ D. frame check sequence field
- E. error correction process field

Q70. A network administrator is adding a new LAN to a branch office. The new LAN must support 4 connected devices. What is the smallest network mask that the network administrator can use for the new network?

- ✓ A. 255.255.255.248
- B. 255.255.255.0
- C. 255.255.255.128
- D. 255.255.255.192

Q71. What service is provided by POP3?

- ✓ A. Retrieves email from the server by downloading the email to the local mail application of the client.
- B. An application that allows real-time chatting among remote users.
- C. Allows remote access to network devices and servers.
- D. Uses encryption to provide secure remote access to network devices and servers.

Q72. What two security solutions are most likely to be used only in a corporate environment? (Choose two.)

- A. antispymware
- ✓ B. virtual private networks
- ✓ C. intrusion prevention systems
- D. strong passwords
- E. antivirus software

Q73. What characteristic describes antivirus software?

✓ **A. applications that protect end devices from becoming infected with malicious software**

- B. a network device that filters access and traffic coming into a network
- C. a tunneling protocol that provides remote users with secure access into the network of an organization
- D. software on a router that filters traffic based on IP addresses or applications

Q74. What mechanism is used by a router to prevent a received IPv4 packet from traveling endlessly on a network?

A. It checks the value of the TTL field and if it is 0, it discards the packet and sends a Destination Unreachable message to the source host.

B. It checks the value of the TTL field and if it is 100, it discards the packet and sends a Destination Unreachable message to the source host.

✓ **C. It decrements the value of the TTL field by 1 and if the result is 0, it discards the packet and sends a Time Exceeded message to the source host.**

D. It increments the value of the TTL field by 1 and if the result is 100, it discards the packet and sends a Parameter Problem message to the source host.

Q75. A client packet is received by a server. The packet has a destination port number of 69. What service is the client requesting?

- A. DNS
- B. DHCP
- C. SMTP

✓ **D. TFTP**

Q76. An administrator defined a local user account with a secret password on router R1 for use with SSH. Which three additional steps are required to configure R1 to accept only encrypted SSH connections? (Choose three.)

A. Configure DNS on the router.

B. Generate two-way pre-shared keys.

✓ **C. Configure the IP domain name on the router.**

✓ **D. Generate the SSH keys.**

✓ **E. Enable inbound vty SSH sessions.**

F. Enable inbound vty Telnet sessions.

Q77. Which two functions are performed at the MAC sublayer of the OSI Data Link Layer to facilitate Ethernet communication? (Choose two.) [implements trailer w/ FCS for error detection; implements process to delimit fields within Ethernet 2 frame]

- A. handles communication between upper layer networking software and Ethernet NIC hardware

✓ **B. implements trailer with frame check sequence for error detection**

C. places information in the Ethernet frame that identifies which network layer protocol is being encapsulated by the frame

✓ **D. implements a process to delimit fields within an Ethernet 2 frame**

E. adds Ethernet control information to network protocol data

Q78. An IPv6 enabled device sends a data packet with the destination address of FF02::2. What is the target of this packet?

A. all IPv6 enabled devices on the local link

B. all IPv6 DHCP servers

C. all IPv6 enabled devices across the network

✓ **D. all IPv6 configured routers on the local link**

Q79. What are the three parts of an IPv6 global unicast address? (Choose three.)

✓ **A. subnet ID**

B. subnet mask

C. broadcast address

✓ **D. global routing prefix**

✓ **E. interface ID**

Q80. A network administrator is designing the layout of a new wireless network. Which three areas of concern should be accounted for when building a wireless network? (Choose three.)

A. extensive cabling

B. mobility options

C. packet collision

✓ **D. interference**

✓ **E. security**

✓ **F. coverage area**

Q81. A new network administrator has been asked to enter a banner message on a Cisco device. What is the fastest way a network administrator could test whether the banner is properly configured?

A. Enter CTRL-Z at the privileged mode prompt.

B. Exit global configuration mode.

C. Power cycle the device.

D. Reboot the device.

✓ **E. Exit privileged EXEC mode and press Enter.**

Q82. What method is used to manage contention-based access on a wireless network?

- A. token passing
- ✓ B. CSMA/CA
- C. priority ordering
- D. CSMA/CD

Q83. What is a function of the data link layer?

- A. provides the formatting of data
- B. provides end-to-end delivery of data between hosts
- C. provides delivery of data between two applications
- ✓ D. provides for the exchange of frames over a common local media

Q84. What is the purpose of the TCP sliding window?

- A. to ensure that segments arrive in order at the destination
- B. to end communication when data transmission is complete
- C. to inform a source to retransmit data from a specific point forward
- ✓ D. to request that a source decrease the rate at which it transmits data

Q85. What characteristic describes spyware?

- A. a network device that filters access and traffic coming into a network
- ✓ B. software that is installed on a user device and collects information about the user
- C. an attack that slows or crashes a device or network service
- D. the use of stolen credentials to access private data

Q86. Which switching method drops frames that fail the FCS check?

- ✓ A. store-and-forward switching
- B. borderless switching
- C. ingress port buffering
- D. cut-through switching

Q87. Which range of link-local addresses can be assigned to an IPv6-enabled interface?

- A. FEC0::/10
- B. FDEE::/7
- ✓ C. FE80::/10
- D. FF00::/8

Q88. What service is provided by FTP?

- A. A basic set of rules for exchanging text, graphic images, sound, video, and other multimedia files on the web.
- B. An application that allows real-time chatting among remote users.
- ✓ C. Allows for data transfers between a client and a file server.
- D. Uses encryption to secure the exchange of text, graphic images, sound, and video on the web.

Q89. A user is attempting to access `http://www.cisco.com/` without success. Which two configuration values must be set on the host to allow this access? (Choose two.)

- ✓ A. DNS server
- B. source port number
- C. HTTP server
- D. source MAC address
- ✓ E. default gateway

Q90. Which two statements accurately describe an advantage or a disadvantage when deploying NAT for IPv4 in a network? (Choose two.)

- A. NAT adds authentication capability to IPv4.
- ✓ B. NAT introduces problems for some applications that require end-to-end connectivity.
- C. NAT will impact negatively on switch performance.
- ✓ D. NAT provides a solution to slow down the IPv4 address depletion.
- E. NAT improves packet handling.
- F. NAT causes routing tables to include more information.

Q91. What would be the interface ID of an IPv6 enabled interface with a MAC address of 1C-6F-65-C2-BD-F8 when the interface ID is generated by using the EUI-64 process?

- A. 0C6F:65FF:FEC2:BDF8
- ✓ B. 1E6F:65FF:FEC2:BDF8
- C. C16F:65FF:FEC2:BDF8
- D. 106F:65FF:FEC2:BDF8

Q92. Refer to the exhibit. PC1 issues an ARP request because it needs to send a packet to PC2. In this scenario, what will happen next?

- A. SW1 will send an ARP reply with the SW1 Fa0/1 MAC address.
- B. SW1 will send an ARP reply with the PC2 MAC address.
- ✓ C. PC2 will send an ARP reply with the PC2 MAC address.
- D. RT1 will send an ARP reply with the RT1 Fa0/0 MAC address.
- E. RT1 will send an ARP reply with the PC2 MAC address.

Q93. What service is provided by BOOTP?

- A. Uses encryption to secure the exchange of text, graphic images, sound, and video on the web.
- B. Allows for data transfers between a client and a file server.
- ✓ C. Legacy application that enables a diskless workstation to discover its own IP address and find a BOOTP server on the network.
- D. A basic set of rules for exchanging text, graphic images, sound, video, and other multimedia files on the web.

Q94. What characteristic describes adware?

- A. a network device that filters access and traffic coming into a network
- ✓ B. software that is installed on a user device and collects information about the user
- C. the use of stolen credentials to access private data
- D. an attack that slows or crashes a device or network service

Q95. When a switch configuration includes a user-defined error threshold on a per-port basis, to which switching method will the switch revert when the error threshold is reached?

- A. cut-through
- ✓ B. store-and-forward
- C. fast-forward
- D. fragment-free

Q96. Match a statement to the related network model. (P2P network: no dedicated server required, client/server roles set per request basis. P2P application: requires specific user interface, background service required.)

- ✓ A. Peer-to-peer NETWORK -> no dedicated server is required
- ✓ B. Peer-to-peer NETWORK -> client and server roles are set on a per request basis
- ✓ C. Peer-to-peer APPLICATION -> requires a specific user interface
- ✓ D. Peer-to-peer APPLICATION -> a background service is required

Q97. What are two primary responsibilities of the Ethernet MAC sublayer? (Choose two.)

- A. error detection
- B. frame delimiting
- ✓ C. accessing the media
- ✓ D. data encapsulation
- E. logical addressing

Q98. Refer to the exhibit. What three facts can be determined from the viewable output of the show ip interface brief command? (Choose three.)

- A. Two physical interfaces have been configured.

- ✓ B. The switch can be remotely managed.
- ✓ C. One device is attached to a physical interface.
- D. Passwords have been configured on the switch.
- E. Two devices are attached to the switch.
- ✓ F. The default SVI has been configured.

Q99. Match each type of frame field to its function. (addressing/error detection/type/frame start)

- ✓ A. addressing -> This field helps to direct the frame toward its destination.
- ✓ B. error detection -> This field checks if the frame has been damaged during the transfer.
- ✓ C. type -> This field is used by the LLC to identify the Layer 3 protocol.
- ✓ D. frame start -> This field identifies the beginning of a frame.

Q100. What is the subnet ID associated with the IPv6 address 2001:DA48:FC5:A4:3D1B::1/64?

- A. 2001:DA48::/64
- B. 2001:DA48:FC5::A4:/64
- ✓ C. 2001:DA48:FC5:A4::/64
- D. 2001::/64

Q101. Match the firewall function to the type of threat protection it provides to the network.

- ✓ A. prevents access by port number -> application filtering
- ✓ B. prevents access based on IP or MAC address -> packet filtering
- ✓ C. prevents unsolicited incoming sessions -> stateful packet inspection
- ✓ D. prevents access to websites -> URL filtering

Q102. Users are reporting longer delays in authentication and in accessing network resources during certain time periods of the week. What kind of information should network engineers check to find out if this situation is part of a normal network behavior?

- A. syslog records and messages
- ✓ B. the network performance baseline
- C. debug output and packet captures
- D. network configuration files

Q103. How does the service password-encryption command enhance password security on Cisco routers and switches?

- A. It requires encrypted passwords to be used when connecting remotely to a router or switch with Telnet.

✓ **B. It encrypts passwords that are stored in router or switch configuration files.**

- C. It requires that a user type encrypted passwords to gain console access to a router or switch.
- D. It encrypts passwords as they are sent across the network.

Q104. Which two statements are correct in a comparison of IPv4 and IPv6 packet headers? (Choose two.)

✓ **A. The Source Address field name from IPv4 is kept in IPv6.**

- B. The Version field from IPv4 is not kept in IPv6.
- C. The Destination Address field is new in IPv6.
- D. The Header Checksum field name from IPv4 is kept in IPv6.

✓ **E. The Time-to-Live field from IPv4 has been replaced by the Hop Limit field in IPv6.**

Q105. A network administrator wants to have the same network mask for all networks at a particular small site (IP phones - 22, PCs - 20, Printers - 2, Scanners - 2). The site uses 192.168.10.0/24. Which single subnet mask would make the most efficient use of the available addresses for the four subnetworks?

- A. 255.255.255.192
- B. 255.255.255.252
- C. 255.255.255.240
- D. 255.255.255.248
- E. 255.255.255.0

✓ **F. 255.255.255.224**

Q106. What characteristic describes identity theft?

✓ **A. the use of stolen credentials to access private data**

- B. software on a router that filters traffic based on IP addresses or applications
- C. software that identifies fast-spreading threats
- D. a tunneling protocol that provides remote users with secure access into the network of an organization

Q107. A network administrator is adding a new LAN to a branch office. The new LAN must support 200 connected devices. What is the smallest network mask that the network administrator can use for the new network?

- A. 255.255.255.240

✓ **B. 255.255.255.0**

- C. 255.255.255.248
- D. 255.255.255.224

Q108. What are three commonly followed standards for constructing and installing cabling? (Choose three.)

A. cost per meter (foot)

✓ B. cable lengths

C. connector color

✓ D. pinouts

✓ E. connector types

F. tensile strength of plastic insulator

Q109. Refer to the exhibit. What is wrong with the displayed termination?

A. The woven copper braid should not have been removed.

B. The wrong type of connector is being used.

✓ C. The untwisted length of each wire is too long.

D. The wires are too thick for the connector that is used.

Q110. Match the characteristic to the category. (IP address vs MAC address)

✓ A. IP address -> contained in the Layer 3 header

✓ B. MAC address -> contained in the Layer 2 header

✓ C. IP address -> separated into a network portion and a unique identifier

✓ D. MAC address -> separated into OUI and a unique identifier

✓ E. IP address -> 32 or 128 bits

✓ F. MAC address -> 48 bits

Q111. A client packet is received by a server. The packet has a destination port number of 143. What service is the client requesting?

✓ A. IMAP

B. FTP

C. SSH

D. Telnet

Q112. What are two characteristics shared by TCP and UDP? (Choose two.)

A. default window size

B. connectionless communication

✓ C. port numbering

D. 3-way handshake

E. ability to carry digitized voice

✓ F. use of checksum

Q113. Refer to the exhibit. Which two network addresses can be assigned to the network containing 10 hosts, within the 10.18.10.0/24 range, wasting the fewest addresses? (Choose two.)

A. 10.18.10.200/28

✓ B. 10.18.10.208/28

C. 10.18.10.240/27

D. 10.18.10.200/27

E. 10.18.10.224/27

✓ F. 10.18.10.224/28

Q114. A client packet is received by a server. The packet has a destination port number of 21. What service is the client requesting?

✓ A. FTP

B. LDAP

C. SLP

D. SNMP

Q115. What attribute of a NIC would place it at the data link layer of the OSI model?

A. attached Ethernet cable

B. IP address

✓ C. MAC address

D. RJ-45 port

E. TCP/IP protocol stack